REPRESENTABLE MORPHISMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1091-1101, following the statement at 1083-1089.

Editorial note. The proof is complete with no omission markers; the displayed chain is the whole argument.

Allowed standard prerequisites (editorial).

- Representable morphism of presheaves. the representability of f: x -> y fixed in the statement, used at proof line 1100. Supplies that the functor w mapping to h_x(w) x_{h_y(w)} h_z(w) is representable for every z -> y'.
- Presheaf fibre products computed objectwise. the displayed expansion at proof lines 1095-1099. Lets h_{x'}(w) be expanded as h_x(w) x_{h_y(w)} h_{y'}(w) via the defining fibre product x' = x times_y y'.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $f : x \to y$ be representable. Let $y' \to y$ be a morphism of $\mathcal{C}$. Then the morphism $x' := x \times_y y' \to y'$ is representable also.*

Proof. Let $z \to y'$ be a morphism. The fibre product $x' \times_{y'} z$ is supposed to represent the functor

$$\begin{aligned} w &\mapsto h_{x'}(w) \times_{h_{y'}(w)} h_z(w) \\ &= (h_x(w) \times_{h_y(w)} h_{y'}(w)) \times_{h_{y'}(w)} h_z(w) \\ &= h_x(w) \times_{h_y(w)} h_z(w) \end{aligned}$$

which is representable by assumption. □

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